Zhao Tang

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SUMMARY

Seeking a full-time software engineer position starting from June 2021.

PhD candidate with strong programming, analytical and designing skills. A wide range of project experience on computer vision, web and mobile development with solid programming skills and computer science knowledge.

EDUCATION

Jun 2012-May 2021 (expected) **Rice University, Houston, TX**

PhD in Chemical and Biomolecular Engineering, GPA 3.63/4.00 Research Advisor: Prof. Matteo Pasquali

Sep 2009-Jun 2012 **Purdue University, West Lafayette, IN**

BSE with honor degree in Chemical Engineering, GPA 3.76/4.00 Research Advisor: Prof. Alex Wei

Courses: Algorithms, Data Structures, Machine Learning, Digital Image Processing, Numerical Methods

WORK EXPERIENCES

Rice University Complex Flow and Complex Fluids Lab Houston, TX

Research Assistant Jun. 2012 – May. 2021

- Developed automated **Matlab** video processing code to track the Brownian motion and super resolution bending of single molecular polymer chain.
- Optimized **Matlab** code and leveraged parallel computing in cluster, achieving a reduction of video processing time from 15 hours to 30 minutes.
- Published 10+ journal articles and conference proceedings

TECHNICAL SKILLS

- Programming language: Java, Matlab, SQL, Python, C, FORTRAN
- Web Mobile Development: Java Servlet, AngularJS, Node.js, HTML & CSS, React, Android
- Databases and Cloud: MySQL, AWS EC2

PROJECT

Job+: A Personalized Job Recommendation Engine (https://github.com/zt5rice/jobplus.git)

- Designed and implemented an interactive web app for users to search and apply available job positions.
- Performed front-end web UI design and implementation using **HTML/CSS/JavaScript**.
- Implemented RESTful APIs using Java servlets, retrieved job descriptions using Github API and stored data in **MySQL**.
- Explored multiple recommendation algorithms and extracted keywords from job descriptions to implement a Content-based algorithm.
- Deployed the service to **AWS EC2**.

Hermes: a Spring and Hibernate based Shopping and Ordering system(https://github.com/zt5rice/jobplus)

- Built a web application based on **Spring MVC** to support item search and listing (dependency injection, inversion of control, REST API etc.).
- Implemented security workflow via in-memory and **JDBC** authentication provided by Spring Security.
- Utilized Hibernate to provide better support of database operations
- Developed a Spring Web Flow to support item ordering.

Starlink: React JS based Starlink Trajectory Visualization (https://github.com/zt5rice/space-x4)

- Setup the Repo by leveraging the React official CLI tool, and use **NPM** to manage project dependencies.
- Design the **layout, component interface** and **data flow** before the implementation.
- Built forms to collect user observation geo information using **Ant Design** component library.
- Fetch nearby satellites information and position prediction data through the **N2YO API(s)**
- Animated selected satellite paths on a world map using **React-Simple-Map** to improve the user friendness.
- Deployed the dashboard to **AWS** for demonstration.

Tinnews: a Tinder-like News App

- Designed the Instagram Flavor News app based on Google Component Architectural MVVM Pattern
- Implemented the bottom bar & page navigation using JetPack navigation component
- Utilized 3rd party CardStackView(RecyclerView) to support swipe gestures for liking/disliking the news
- Built the Room Database with LiveData & ViewModel to support local cache and offline model
- Integrated Retrofit and LiveData to pull the latest news data from a RESTFUL endpoint (newsapi.org)